Siddesh Patil

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Summary

Computer Science & Engineering (Data Science) graduate with Honours in AI/ML; published IEEE researcher in AIoT-based precision agriculture. Built and evaluated production ML systems, 92.86%-accurate rainfall prediction, 97.2%-accurate crop classification, and hybrid RAG pipelines seeking ML/AI Engineer or Data Scientist roles.

Projects

SoilStream | *Python, scikit-learn, TensorFlow, ResNet50, ESP32, Flask, Firebase* | GitHub

- Predicted regional rainfall with 92.86% accuracy using an SVR model, outperforming LightGBM and GRU baselines enabling smarter irrigation timing during monsoon season.
- Built a ResNet50 crop-classification model (97.2% accuracy) integrated into an automated irrigation controller, reducing waterlogging risk and manual monitoring effort.
- Designed a full IoT-to-cloud pipeline (ESP32 + BME280 sensors → Firebase → mobile app) for real-time field monitoring, remote pump control, and crop disease diagnosis via the Kindwise API.

VeriDoc | *Python, FAISS, BM25, Sentence-Transformers, Mistral, MLflow, Gradio* | GitHub

- Built a hybrid retrieval pipeline (dense FAISS embeddings + sparse BM25) over local PDFs, improving answer relevance over single-method retrieval before generating cited answers with an LLM (Mistral).
- Ran exploratory analysis and evaluation on retrieval outputs using the RAGAS framework, tracking faithfulness and relevancy metrics across experiment runs with MLflow bringing a data-driven approach to LLM output quality.

Sidekik - A freelancing Platform | *Flask, Python, BERT, FAISS, Elasticsearch, Supabase, PostgreSQL* | GitHub

- Built a BERT + FAISS semantic matching engine to rank freelancer-project fit, improving match relevance over keyword-only search across a corpus of freelancer profiles and project records.
- Added semantic search via Elasticsearch and integrated the Adzuna API for real-time skill-demand analytics.
- Shipped a full-stack platform end-to-end (Flask APIs, Supabase backend, marketplace UI) – from data pipeline to deployed product.

Publications & Intellectual Property

Research Paper - IEEE CONIT 2026 (Published) Paper ID: 960
A Solar-Powered AIoT-Based System for Intelligent Soil Monitoring and Automated Irrigation Control

Patent Filed - Government of India App. No: 202621042187
Smart Solar-Powered AIoT System for Precision Agriculture (SoilStream)

Education

University of Mumbai 2022 - 2026
B.E. Computer Science and Engineering (Data Science) | *Honours in AI/ML*

Technical Skills

Languages: Python, SQL, Java, C
AI/ML: NLP, RAG, Recommendation Systems, SVR, CNNs, LSTM, EDA
Frameworks: Flask, Pandas, NumPy, Matplotlib, Sentence-Transformers, FAISS, MLflow, Jupyter Notebook
Cloud & Tools: Git, GitHub, Firebase, Supabase, Linux, AWS (EC2, S3, IAM)

Certifications

Generative AI | *Google Cloud Virtual Internship (AICTE)*

Google AI/ML | *Google Cloud Virtual Internship (AICTE)*

Data Science Methodology | *IBM Developer Skills Network*